

Article

Comparative Analysis of AR-HUDs Crash Warning Icon Designs: An Eye-Tracking Study Using 360° Panoramic Driving Simulation

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Abstract: Augmented Reality Head-Up Displays (AR-HUDs) enhance driver perception and safety, yet optimal hazard warning design remains unclear. This study examines three AR-HUD crash warning icon types (BD, BR, BW) across various turning scenarios. Using a 360-degree video-based driving simulation with 36 participants, eye-tracking metrics were collected. Results show BW icons, dynamically linked to hazards, significantly improve drivers' pedestrian risk awareness and visual attention allocation compared to BD and BR systems. BW consistently demonstrated longer gaze duration, higher fixation counts, and shorter time to first fixation across all turns. BD and BR icons were more susceptible to lane changes, potentially diverting attention from hazards. Findings suggest prioritizing dynamic tracking warning icons over fixed-position alternatives to minimize visual competition and distraction, providing crucial insights for AR-HUD optimization in automated vehicles.

Keywords: AR-HUDs; crash warning icon systems; driving safety; eye-tracking; visual attention



Citation: Wu, Z.; Liang, Y.; Liu, G.; Ai, X. Comparative Analysis of AR-HUDs Crash Warning Icon Designs: An Eye-Tracking Study Using 360° Panoramic Driving Simulation. *Sustainability* **2024**, *16*, 9167. <https://doi.org/10.3390/su16219167>

Academic Editors: Chunlu Liu, Mirosław J. Skibniewski, Hui Liu and Yu Lei

Received: 1 July 2024

Revised: 25 September 2024

Accepted: 14 October 2024

Published: 22 October 2024



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1. Introduction

1.1. Background on Head-Up Displays (HUDs) and Augmented Reality (AR)

Advanced driver assistance systems (ADAS) are increasingly integrated into modern vehicles, providing drivers with additional sources of information. As a component of ADAS, head-up display (HUD) technology projects crucial visual data onto a transparent screen within the driver's field of view, enabling access to driving information without diverting attention from the road [1]. This innovative technology has demonstrated significant benefits across multiple aspects of driving, including enhanced performance [2], improved situational awareness [3], reduced task response times [4], and optimized driver workload management [5,6].

Recent advancements in augmented reality (AR) technology have further propelled HUD development, overcoming previous limitations in brightness, resolution, field of view, and multi-focal plane capabilities [7]. These improvements have led to the emergence of AR-HUD technology. AR-HUDs seamlessly integrates information into the driver's field of view, merging displayed data with the real driving environment [8]. These systems show promise in enhancing drivers' situational awareness and trust in vehicle systems [9,10].

1.2. Challenges in AR-HUD Crash Warning Icon Systems Design

AR-HUD icon systems aim to reduce vehicular collision incidents by promptly issuing potential risk notifications to drivers which can be categorized into two main types: screen-fixed icons and world-relative icons [11]. Screen-fixed icons display information at fixed positions on the HUD interface, a method commonly employed in many current HUD systems [12]. Many driving studies exploring AR-HUD icon systems have utilized screen-fixed display schemes, as evidenced in studies [13–19]. With advancements in tracking and spatial localization technologies, AR icons can now be associated with the real driving

environment. This integration creates the impression that the icons and the road are unified, resulting in world-relative icons. These icons enhance spatial localization and environmental awareness capabilities. Compared to screen-fixed icons, world-relative icons can more rapidly draw drivers' attention to potential hazards, offering improved safety benefits.

Recent research has challenged the presumed superiority of world-relative icons in AR-HUD systems. Merenda et al. [20] conducted a comparative study on the effects of screen-fixed and world-relative navigation icons on driver performance and visual attention. Their findings revealed that world-relative navigation icons may lead to extended task response times and potentially compete for attention resources typically allocated to real-world visual stimuli. Further investigation by Smith et al. [8] examined the impact of AR navigation icons' spatial position, turning direction, and traffic presence on driver behavior and preferences. Contrary to expectations, their results demonstrated a driver preference for screen-fixed icons over world-relative alternatives. These findings collectively suggest that integrating head-up display (HUD) system information directly into the driving environment may not be the optimal solution, particularly in complex urban settings. The potential cognitive load and attention competition introduced by world-relative icons could outweigh their intended benefits, necessitating careful consideration in the design and implementation of AR-HUD systems for automotive applications.

Early research by Horrey and Wickens [21] demonstrated a significant increase in traffic accident risk when drivers' gaze deviates from the road for more than two seconds. This finding underscores the critical need for drivers to swiftly detect, comprehend, and respond to vehicle crash warnings by redirecting attention to the road, thereby facilitating timely reactions to potential hazards and mitigating accident risks. Despite the importance of effective crash warning systems, current research has insufficiently addressed key parameters of Head-Up Display (HUD) crash warning systems, particularly regarding the types and display locations of such information. These factors play a crucial role in driving performance when hazards are present in the driving environment. Improperly designed warning system attributes may have detrimental effects on driver safety. Drivers may fail to detect dangerous objects in the driving environment promptly, or experience cognitive tunneling—a phenomenon where attention becomes fixated on the HUD-displayed crash warning, preventing timely reorientation to the driving environment [14,22].

1.3. Research Gap and Study Objectives

Despite the general presumption that world-relative icons offer advantages in AR-HUDs, they may not be suitable in all scenarios, particularly when warning drivers about dynamically moving pedestrians. This limitation highlights a critical gap in our understanding of AR crash warning icon systems for automotive applications. Consequently, further research is necessary to compare the impact mechanisms of world-relative icons and Screen-fixed icons in AR crash warning icon systems on drivers' visual attention during dynamic pedestrian warning processes. This investigation aims to determine the optimal application method for this specific situation.

To refine our understanding, we have further categorized AR-HUD crash warning icons into three distinct types (Figure 1): Based on Driver's view (BD): Presents an icon directly in the driver's line of sight when a hazardous pedestrian is detected, alerting them to the danger on the road. Based on Road information (BR): Projects an icon onto the road surface ahead of the vehicle upon detecting a hazard. Based on Warning information (BW): Displays the icon at the location of the hazardous pedestrian and moves along with the pedestrian to continuously indicate the danger.

By examining the influence of these three warning types on drivers' visual attention, this research will contribute to the development of more nuanced and context-aware AR interfaces for advanced driver assistance systems. This approach offers several advantages: Comprehensive analysis: Considering multiple warning types allows for a more holistic understanding of how different AR-HUD presentations affect driver attention. Contextual

relevance: Each warning type addresses a specific aspect of the driving environment, enabling targeted analysis of driver responses. Optimization potential: Understanding the unique impacts of each warning type can lead to more effective and intuitive AR-HUD designs. Safety enhancement: Tailoring warning presentations to specific scenarios (driver perspective, road conditions, or dynamic targets) may significantly improve driver awareness and reaction times.

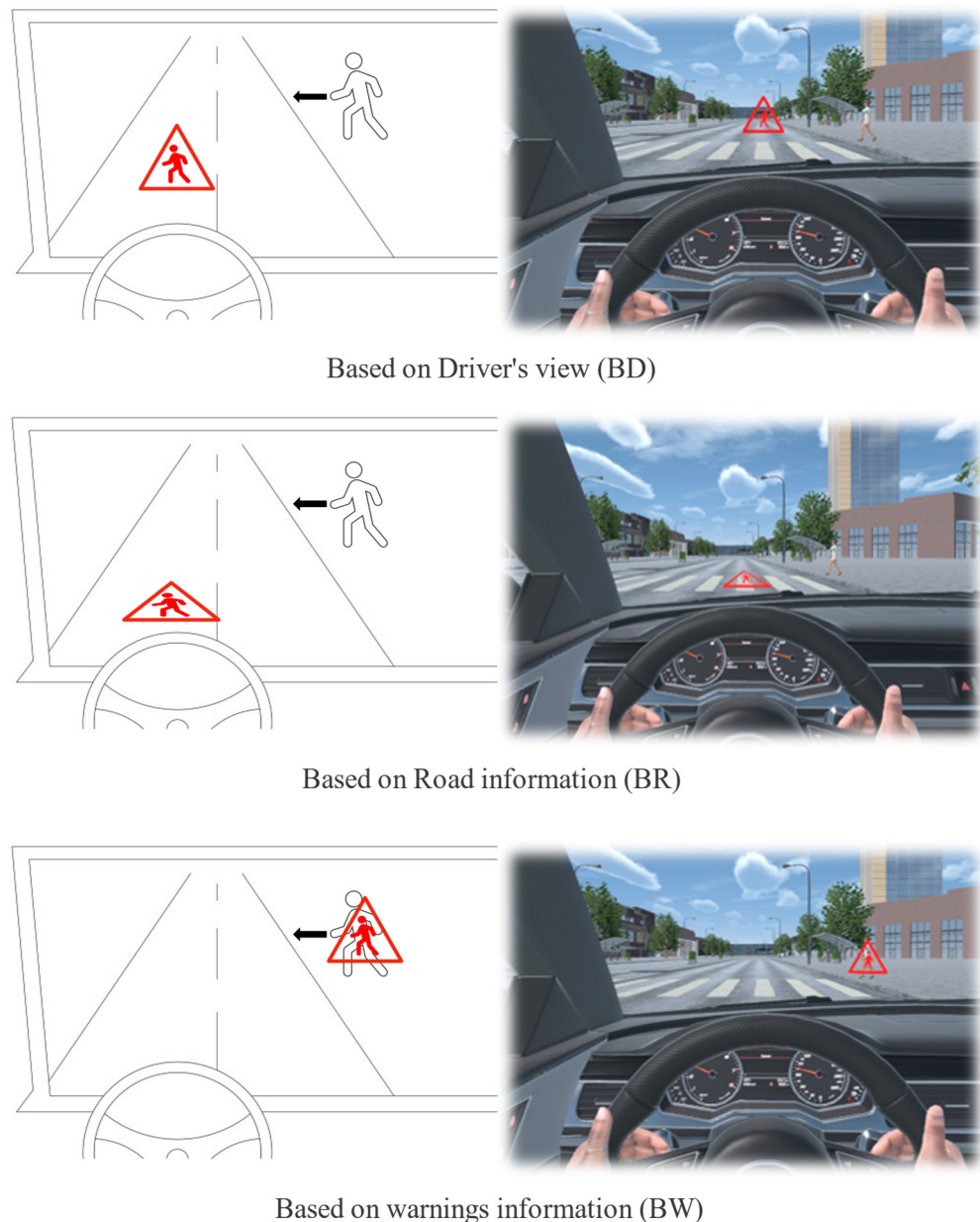


Figure 1. Three kinds of AR-HUD crash warning icon types.

This study aims to evaluate the efficacy of diverse augmented reality (AR) crash warning system types in mitigating pedestrian collision risks. By comparing these display strategies, we seek to determine their relative effectiveness in enhancing driver awareness and response to potential pedestrian collisions in urban driving environments. Based on a comprehensive synthesis of prior research conclusions, this study proposes the following hypotheses:

H1. *The effectiveness of AR crash warning icon systems in enhancing drivers' perception of potential hazards will differ significantly among three warning display types:*

- (a) *Based on driver (BD): AR crash warning icon systems displayed directly in the driver's line of sight will improve hazard detection compared to no-warning conditions.*
- (b) *Based on road (BR): AR crash warning icon systems projected onto the road surface in front of the vehicle will enhance hazard perception compared to no-warning conditions.*
- (c) *Based on warning (BW): AR crash warning icon systems displayed at the location of hazardous pedestrians and moving with them will most effectively facilitate hazard identification compared to BD and BR displays.*

H2. *The impact of different turning lane types on drivers' visual behavior and hazard perception will exhibit significant variations:*

- (a) *In straight lanes, drivers will demonstrate the highest level of hazard perception across all three AR-HUD crash warning types.*
- (b) *In left-turn lanes, drivers' hazard perception will slightly decrease compared to straight lanes, but the effectiveness of AR crash warnings (especially Based on Warning, BW) will remain significant.*
- (c) *In right-turn lanes, drivers will exhibit the lowest level of hazard perception, yet AR crash warnings (particularly BW) will still provide substantial improvements in hazard detection.*

To test this hypothesis, we conducted an experiment to acquire drivers' eye movement data during simulated driving based on 360-degree video demonstrations. This approach allows us to understand the dynamic visual attention mechanisms of drivers. The utilization of 360-degree video-based simulated driving environments provides a comprehensive and immersive platform for examining drivers' visual behaviors. By capturing eye movement data within this controlled yet realistic setting, we can gain valuable insights into the dynamic nature of drivers' visual attention allocation.

Eye-tracking technology is highly valuable in studying drivers' visual attention patterns. Previous research has utilized this technology to analyze the impact of Advanced Driver Assistance Systems (ADAS) on driving behavior. For example, Falkmer and Gregersen [23] used eye-tracking data to compare visual search strategies between experienced and inexperienced drivers, revealing that the latter focused more on in-car objects and traffic cues. Li et al. [24] assessed the usability and safety effects of different in-vehicle display systems across various speeds, measuring average gaze duration, number of gazes, and blink rate. Hajiseyedjavadi et al. [25] investigated how warning icons influence young drivers' hazard perception and visual attention. Similarly, Yang et al. [26] studied the impact of warning icon placements on driving performance and visual patterns in different traffic accident scenarios. By analyzing eye-tracking data during alert notifications, researchers can gain insights into the visual attention mechanisms underlying driving behavior, thus improving the design and safety of warning systems.

2. Methods

2.1. Warning Icon Types

This study explores the influence of three types of warning icons on drivers' visual behavior in various road-turning scenarios during vehicular accidents. These icons are depicted in red (RGB 255, 0, 0) due to their universally recognized association with caution and danger, as evidenced in the existing literature. The design of the pedestrian warning icons aligns with the style used in the research conducted by Winkler et al. [16].

2.2. Virtual Driving Scenarios and Human-Vehicle Event Allocation

This study employs Unity 3D (version 2019.1.19) to construct a driving simulation environment miming a two-way, four-lane urban road setting. Each lane is 3.5 m wide with a speed limit of 60 km/h, and a greenbelt separates the opposing lanes to prevent incorrect driving. The simulation includes urban elements like building infrastructure, green spaces, and traffic signs to enhance realism. The vehicle-pedestrian interaction process is depicted in Figure 2. When a vehicle nears a pedestrian crossing within 60 m, the pedestrian begins

to cross at a speed of 1.5 m/s. If the time to collision (TTC) drops below 3 s, different pedestrian warning icons are displayed on the vehicle's windshield to alert the driver of potential hazards. Drivers encounter pedestrian crossing scenarios while going straight, turning left, and turning right at intersections, as illustrated in Figure 3.

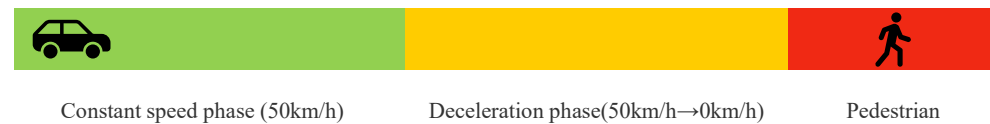


Figure 2. Vehicle-pedestrian interactions process.

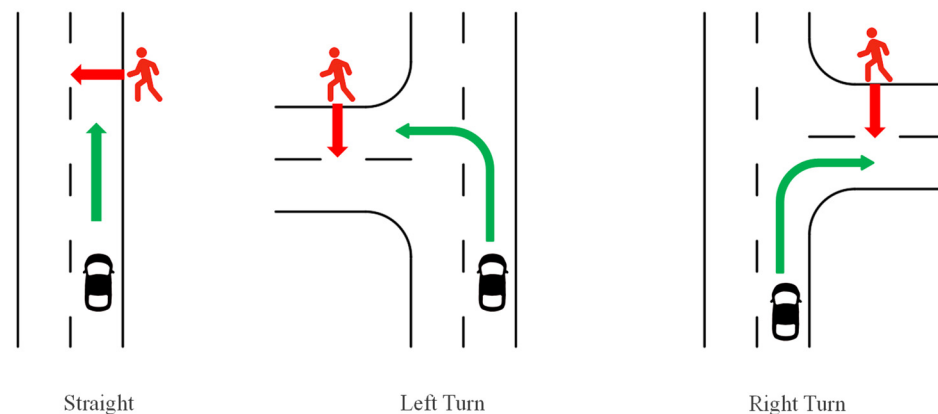


Figure 3. Three key steering lanes.

2.3. Participants

The impact of drivers' age and driving experience on driving performance has been well documented in previous studies [27,28]. To ensure the validity of the results and minimize the influence of variables such as age, driving experience, emotional state, and cognitive abilities, a consistent driving simulation protocol was employed in this study. Thirty-nine individuals participated, but three had to withdraw due to severe dizziness and discomfort, resulting in an effective sample size of 36, with an average age of 23.7 years ($SD = 2.315$). All participants possessed valid Chinese driver's licenses, had an average of 3.2 years of driving experience ($SD = 1.869$), and had normal or corrected-to-normal vision. Of the participants, 22.2% had some familiarity with the head-up display system, while 77.8% were entirely unfamiliar with it, and 26.67% had prior experience with driving simulators. The participants were randomly assigned to three groups and received compensation for their time and participation. This study was approved by the Department of Industrial Design at Huaqiao University.

2.4. Apparatus

In recent years, developers have integrated eye-tracking technology into virtual reality (VR) systems using head-mounted displays (HMDs) as advancements in video-based eye-tracking have advanced [29]. This study employed the HTC VIVE Pro headset along with the HTC VIVE Pro Eye head-mounted eye-tracking system. The experimental setup included a workstation connected to the device for playing driving simulation videos and gathering eye-tracking data. The HTC VIVE Pro features dual 3.5-inch OLED displays with a resolution of 1440×1600 per eye, a refresh rate of 90 Hz, and a field of view of 110 degrees. The HTC VIVE Pro Eye can record eye movement data at 120 Hz with an eye-tracking accuracy of $0.5\text{--}1.1^\circ$ and a visual angle of 110 degrees. The workstation is equipped with Tobii Pro Lab software (1.152) for collecting and analyzing the driver's eye movement data. The specific device configuration and connections are depicted in Figure 4.



Figure 4. Experimental setup for AR-HUD simulation and eye-tracking data collection.

2.5. Experimental Design

This study employs a mixed experimental design, with warning icon type as a between-subjects factor and turning lane as a within-subjects factor. Participants were randomly assigned to one of three AR warning icon groups (based on driver's view (BD), based on road information (BR), and based on warnings information (BW)). Each participant completed driving tasks in three turning scenarios: (a) left turn; (b) straight ahead; (c) right turn. To control for potential order effects, the presentation sequence of these three turning scenarios was randomized across participants.

The experiment's dependent variables related to driver visual attention distribution include First Fixation Time, Total Fixation Duration, Number of Fixations, and Omission Errors. First Fixation Time is the duration between the appearance of the warning icon and the driver's initial focus on the hazardous pedestrian. Total Fixation Duration is the overall time the driver spends observing the hazardous pedestrian during the interaction. Number of Fixations indicates how many times the driver looks at the hazardous pedestrian. Omission Errors occur when the driver fails to notice the warning icon, with an omission being recorded if the driver's gaze does not enter the defined areas of interest associated with the warning icon or the hazardous pedestrian (Figure 5).



(a)AOI division based on driver's information

Figure 5. Cont.



(b) AOI division based on driving road information



(c) AOI division based on hazardous object information

Figure 5. AOIs division of three different groups of AR-HUDs crash warning types.

2.6. Procedure

Study Intake. The initial phase of the study involves preliminary data collection from participants. A basic questionnaire gathers essential information about the subjects, including gender, age, driving experience, familiarity with HUD systems, and prior experience with driving simulators. This preparatory stage provides crucial background information for subsequent experimental activities.

Familiarization Phase. In the subsequent phase, researchers introduce participants to the warning functions of the HUD system and provide detailed explanations of the driving simulation experiment procedures and tasks. The research team assists participants in properly wearing and calibrating the HTC VIVE virtual reality eye-tracking device, adjusting the equipment to ensure the collection of valid eye movement data. Participants then undergo a 2-km driving simulation training session to familiarize themselves with the virtual urban road environment and human-vehicle interaction event processes. This training phase contributes to enhancing the quality of experimental data.

Experimental Phase. During the formal experimental phase, participants experience three driving video segments containing potential hazardous events. Brief rest periods are provided between each video segment, with the order of presentation randomized. Subjects are required to maintain a vehicle speed of 60 km/h before the occurrence of hazardous events in the videos. Upon detecting a potential danger ahead, the vehicle automatically activates its collision warning system, initiates automatic braking, and displays a warning icon on the windshield. Drivers must immediately locate the position of the hazardous object in the scene after the icon appears.

The division into these three stages is crucial for clearly describing the entire experimental process, ensuring that participants are adequately prepared and informed at each stage, and facilitating subsequent data analysis and interpretation. This detailed breakdown of the experimental process is essential for enhancing the efficiency and accuracy of the results.

3. Results

Eye-tracking data were collected for various AR crash warning icon systems across three distinct turning lanes. IBM SPSS 24.0 was used for data analysis. The independent variables examined were the warning icon type and the vehicle's turning lane, while the dependent variables included total fixation duration within the areas of interest (AOIs), number of fixations, and first fixation time. A two-factor analysis of variance (ANOVA) was utilized to analyze the eye-tracking data, with statistical analysis conducted to assess the impact of different warning icon types and turning lanes on drivers' visual attention.

3.1. Total Duration of Fixations

Based on the data processing results, the study found a significant impact on the total fixation duration within the AOIs by the interaction between the type of AR warning icon and the vehicle's turning lane. Therefore, separate effect tests must be conducted for the AR crash warning icon systems and the vehicle's turning lanes. When analyzing the individual effects of the turning lanes, it is crucial to compare the differences in total fixation duration across the three turning lanes under varying AR crash warning icon systems. The results of the within-subject effects test can be seen in Table 1, while the pairwise comparisons between different turning lanes are detailed in Table 2.

Table 1. The results of the within-subject effects test on the impact of turning lane on Total Fixation Duration.

Type	Lane	Mean (s)	SE	F	<i>p</i>
BD	left-turn	3241.000	240.279	5.453	0.012
	straight	3043.667	304.813		
	right-turn	2192.000	450.529		
BR	left-turn	3909.417	327.339	9.961	0.001
	straight	3009.167	265.893		
	right-turn	2253.500	280.743		
BW	left-turn	4742.083	467.588	0.901	0.420
	straight	4879.083	237.148		
	right-turn	5467.167	553.418		

Notes. Significant *p* values are highlighted in bold.

Table 2. Pairwise Comparison of the Effects of Turning Lanes on Total Fixation Duration.

Group	Lane		Mean	SE	<i>p</i>	95% CI	
BD	left-turn	straight	197.333	238.556	1.000	−475.403	870.069
		right-turn	1049.000	423.819	0.093	−146.184	2244.184
	straight	right-turn	851.667	324.687	0.071	−63.962	1767.295
BR	left-turn	straight	900.250	230.598	0.007	249.957	1550.543
		right-turn	1655.917	439.417	0.009	416.746	2895.087
	straight	right-turn	755.667	409.527	0.276	−399.213	1910.546
BW	left-turn	straight	−137.000	445.771	1.000	−1394.089	1120.089
		right-turn	−725.083	674.275	0.916	−2626.562	1176.396
	straight	right-turn	−588.083	578.292	0.993	−2218.886	1042.719

Notes. Significant *p* values are highlighted in bold.

Figure 6a highlights drivers' total fixation duration variations based on different turning lanes and three types of AR crash warning icon systems. The BD group shows a significant interaction between the type of AR warning icon and the turning lane, indicating a mutual influence between the two factors. However, within the same AR warning icon type, no notable differences in fixation duration are observed across different turning lanes in the BD group. This suggests that the impact of turning lanes on fixation duration is absent in this group. Therefore, in the BD group, the type of AR warning icon has a more substantial influence on drivers' attention allocation.

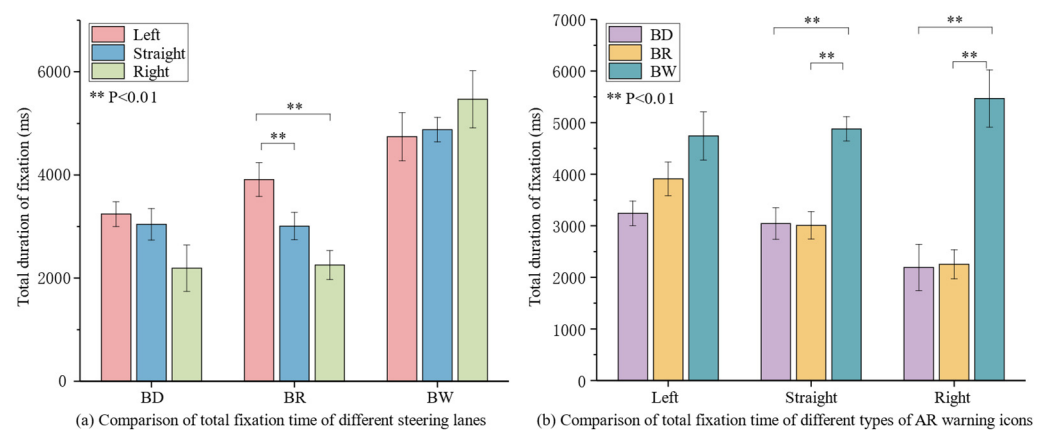


Figure 6. Comparative analysis of total fixation time across different steering lanes and AR crash warning icon systems.

In the BR group, the impact of the turning lane on total fixation duration was statistically significant. There was a notable difference in total fixation time between the left-turn lane and the through lane, as well as between the left-turn lane and the right-turn lane. This suggests that drivers focus more on warnings when in the left-turn lane than other lanes. However, no significant difference was observed between the through and right-turn lanes, indicating similar visual attention demands for these two types of lanes.

In the study conducted by the BW group, it was determined that the influence of the turning lane on total fixation duration was not statistically significant. This suggests that the turning lane does not impact how long drivers focus on displayed warnings within this group. Analyzing the individual effects of AR crash warning icon systems involves comparing the total fixation durations for the three different AR icon types across various turning lanes. The results of the within-subject effects tests can be seen in Table 3, while the pairwise comparisons of the total fixation duration among different AR crash warning icon systems are presented in Table 4. The distribution of total fixation durations for the three types of AR crash warning icon systems across different turning lanes is visually represented in Figure 6b. This visualization aids in evaluating the influence of AR icon type on driver attention in different driving scenarios, highlighting how each type may contribute to hazard perception and response in varying traffic conditions.

Table 3. The results of the within-subject effects test regarding the impact of AR crash warning icon systems on total fixation duration revealed significant differences.

Lane	Group	Mean (ms)	SE	F	<i>p</i>
left-turn	BD	3241.000	240.279	4.167	0.029
	BR	3909.417	327.339		
	BW	4742.083	467.588		
straight	BD	3043.667	304.813	16.208	<0.001
	BR	3009.167	265.893		
	BW	4879.083	237.148		
right-turn	BD	2192.000	450.529	14.800	<0.001
	BR	2253.500	280.743		
	BW	5467.167	553.418		

Table 4. The pairwise comparison results for the impact of different AR crash warning icon systems on total fixation duration.

Lane	Group		MD	SE	<i>p</i>	95% CI	
left-turn	BD	BR	−668.417	431.387	0.449	−1884.943	548.110
		BW	−1501.083	576.991	0.074	−3128.218	126.051
	BR	BW	−832.667	543.391	0.461	−2365.049	699.716
straight	BD	BR	34.500	410.216	1.000	−1122.323	1191.323
		BW	−1835.417	277.388	<0.001	−2617.659	−1053.174
	BR	BW	−1869.917	422.408	0.003	−3061.121	−678.712
right-turn	BD	BR	−61.500	640.938	1.000	−1868.966	1745.966
		BW	−3275.167	799.462	0.005	−5529.678	−1020.656
	BR	BW	−3213.667	610.689	0.001	−4935.830	−1491.503

Notes. Significant *p* values are highlighted in bold.

The study results reveal statistically significant effects of AR crash warning icon systems on total gaze time across different road conditions. For left-turn scenarios, no statistically significant differences were observed among the BD, BR, and BW groups. However, in straight-ahead and right-turn road conditions, the BW group demonstrated significantly different total gaze times compared to both the BD and BR groups, while no statistically significant difference was found between the BD and BR groups. These findings suggest that the BW type of AR warning icon exhibits more pronounced effectiveness in enhancing driver attention.

3.2. Number of Fixations

Further analysis reveals a significant interaction effect between AR crash warning icon systems and vehicle turning lanes on the number of fixation points within AOIs. Consequently, separate effect tests were conducted for the two within-subject factors (icon type and turning lane). The individual effect test for turning lanes involved comparing the differences in the number of fixation points among the three turning lanes under different types of AR crash warning icon systems. Table 5 presents the detailed results of the within-subject effects test, while Table 6 provides pairwise comparisons between different turning lanes.

The fixation points for drivers under three types of AR crash warning icon systems across different turning lanes are illustrated in Figure 7a. The impact of turning lane on fixation points is not statistically significant within the BD group. Conversely, within the BR group, this impact is significant. There is no significant difference between the left-turn and straight lanes. However, the disparity between left-turn and right-turn lanes is statistically significant ($p = 0.001$). Moreover, the difference between straight and right-turn lanes is not significant. Lastly, within the BW group, the influence of turning lane on fixation points is not statistically significant.

Table 5. The results of the within-subject effects test examining how turning lanes influence the total number of fixations points.

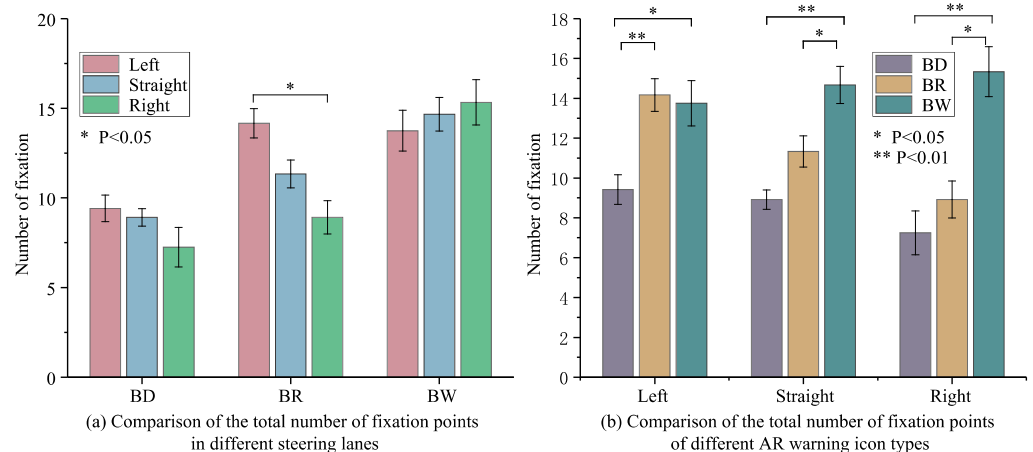
Group	Lane	MD	SE	F	<i>p</i>
BD	L	9.417	0.743	1.518	0.241
	S	8.917	0.484		
	R	7.250	1.102		
BR	L	14.167	0.815	9.042	0.001
	S	11.333	0.782		
	R	8.917	0.925		
BW	L	13.750	1.136	0.731	0.493
	S	14.667	0.932		
	R	15.333	1.257		

Notes. Significant *p* values are highlighted in bold.

Table 6. Results of the pairwise comparison of the total number of fixation points for different lane types.

Type	Lane		MD	SE	<i>p</i>	95% CI	
BD	L	S	0.500	0.634	1.000	−1.287	2.287
		R	2.167	1.669	0.662	−2.540	6.873
	S	R	1.667	1.378	0.756	−2.219	5.553
BR	L	S	2.833	1.230	0.125	−0.635	6.302
		R	5.250	1.067	0.001	2.241	8.259
	S	R	2.417	1.390	0.330	−1.502	6.336
BW	L	S	−0.917	1.417	1.000	−4.912	3.078
		R	−1.583	1.459	0.903	−5.697	2.531
	S	R	−0.667	1.025	1.000	−3.557	2.224

Notes. Significant *p* values are highlighted in bold.

**Figure 7.** Comparative analysis of total number of fixation points across lanes and different AR crash warning icon systems.

To examine the individual effects of icon types, the study compared the differences in the number of fixation points among the three AR crash warning icon systems across different turning lanes. Table 7 presents the results of the within-subject effects test, and Table 8 shows the pairwise comparisons between different turning lanes.

Table 7. The results of the within-subject effects test on the impact of AR crash warning icon systems on the total number of fixation points.

Lane	Type	MD	SE	F	<i>p</i>
L	BD	9.417	0.743	9.823	0.001
	BR	14.167	0.815		
	BW	13.750	1.136		
S	BD	8.917	0.484	17.473	<0.001
	BR	11.333	0.782		
	BW	14.667	0.932		
R	BD	7.250	1.102	13.205	<0.001
	BR	8.917	0.925		
	BW	15.333	1.257		

Table 8. The pairwise comparison results of the effects of AR crash warning icon systems on the total number of fixation points.

Lane	Group		MD	SE	<i>p</i>	95% CI	
L	BD	BR	−4.750	0.922	0.001	−7.351	−2.149
		BW	−4.333	1.269	0.017	−7.913	−0.754
	BR	BW	0.417	1.328	1.000	−3.329	4.163
S	BD	BR	−2.417	1.069	0.135	−5.432	0.599
		BW	−5.750	0.827	<0.001	−8.082	−3.418
	BR	BW	−3.333	1.018	0.022	−6.203	−0.464
R	BD	BR	−2.417	1.069	0.135	−5.432	0.599
		BW	−5.750	0.827	<0.001	−8.082	−3.418
	BR	BW	−3.333	1.018	0.022	−6.203	−0.464

Notes. Significant *p* values are highlighted in bold.

Figure 7b illustrates the impact of the three AR crash warning icon systems on drivers' fixation points across different turning lanes. For left-turn lanes, AR icon types showed statistically significant effects on fixation points. Significant differences were observed between the BD and BR groups, as well as between the BD and BW groups, while the difference between BR and BW groups was not significant. In straight lanes, AR icon types also demonstrated significant effects on fixation points. No significant difference was found between BD and BR groups, but significant differences were observed between BD and BW groups, as well as between BR and BW groups. Similarly, in right-turn lanes, AR icon types significantly influenced the number of fixation points. The difference between BD and BR groups was not significant, while significant differences were found between BD and BW groups, as well as between BR and BW groups.

3.3. Time to First Fixation

The interaction effect analysis reveals a statistically significant impact of the interaction between AR crash warning icon systems and vehicle turning lane on the time to first fixation within the AOI regions. Consequently, separate examinations of the individual effects of AR crash warning icon systems and vehicle turning lane on the time to first fixation were conducted. To assess the individual effect of turning lanes, the study compared differences in time to first fixation among the three turning lanes under different AR crash warning icon systems. Table 9 presents the results of the within-subject effects test, while Table 10 shows the pairwise comparisons between different turning lanes.

Table 9. Results of the within-subject effects test on the influence of lane on first fixation time.

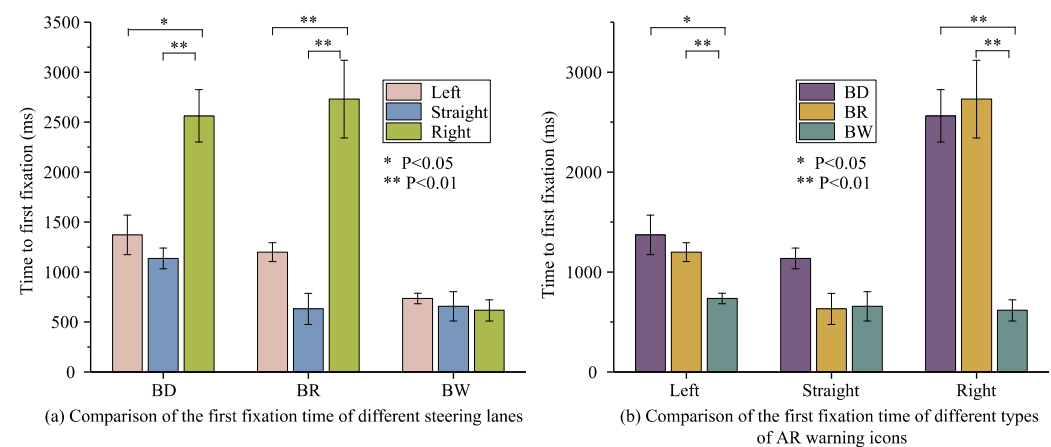
Type	Lane	MD	SE	F	p
BD	L	1371.167	197.780	16.308	<0.001
	S	1136.250	103.138		
	R	2562.583	261.983		
BR	L	1198.917	93.630	22.656	<0.001
	S	630.833	155.850		
	R	2729.917	387.982		
BW	L	735.667	52.660	0.273	0.764
	S	656.750	146.791		
	R	616.667	106.372		

Table 10. Pairwise comparison results of the effects of lane on initial fixation time.

Type	Lane		MD	SE	p	95% CI for Difference	
BD	L	S	234.917	191.096	0.734	−303.980	773.813
		R	−1191.417	349.724	0.018	−2177.649	−205.184
	S	R	−1426.333	237.398	<0.001	−2095.803	−756.864
BR	L	S	568.083	185.324	0.032	45.465	1090.702
		R	−1531.000	349.499	0.003	−2516.599	−545.401
	S	R	−2099.083	394.581	0.001	−3211.814	−986.352
BW	L	S	78.917	169.399	1.000	−398.794	556.627
		R	119.000	99.353	0.769	−161.179	399.179
	S	R	40.083	205.088	1.000	−538.271	618.438

Notes. Significant *p* values are highlighted in bold.

Figure 8a illustrates the performance of initial fixation durations with three types of AR crash warning icon systems across various turning lanes. Turning lanes significantly impact initial fixation durations within the BD group, showing no significant difference between left-turn and straight lanes but significant differences between left-turn and right-turn lanes, and between straight and right-turn lanes. The BR group also experiences significant differences in initial fixation durations based on turning lanes, with significant differences between left-turn and straight lanes, left-turn and right-turn lanes, and straight and right-turn lanes. On the other hand, the BW group does not show a significant effect of turning lanes on initial fixation durations.

**Figure 8.** Comprehensive analysis of first fixation times across lanes and different AR crash warning icon systems.

To evaluate the individual effects of icon types, the study compared the differences in time to first fixation among the three AR crash warning icon systems across different

turning lanes. Table 11 presents the results of the within-subject effects test, and Table 12 shows the pairwise comparisons between different turning lanes.

Table 11. The within-subjects effect test results for the impact of AR crash warning icon systems on first fixation time.

Lane	Type	MD	SE	F	<i>p</i>
L	BD	1371.167	197.780	5.793	0.010
	BR	1198.917	93.630		
	BW	735.667	52.660		
S	BD	1136.250	103.138	5.208	0.014
	BR	630.833	155.850		
	BW	656.750	146.791		
R	BD	2562.583	261.983	18.326	<0.001
	BR	2729.917	387.982		
	BW	616.667	106.372		

Table 12. The pairwise comparison results for the impact of AR crash warning icon systems on first fixation time.

Lane	Type		MD	SE	<i>p</i>	95% CI for Difference	
L	BD	BR	172.250	252.450	1.000	−539.669	884.169
		BW	635.500	207.404	0.032	50.612	1220.388
	BR	BW	463.250	71.710	<0.001	261.027	665.473
S	BD	BR	172.250	252.450	1.000	−539.669	884.169
		BW	635.500	207.404	0.032	50.612	1220.388
	BR	BW	463.250	71.710	<0.001	261.027	665.473
R	BD	BR	−167.333	411.081	1.000	−1326.595	991.928
		BW	1945.917	285.751	<0.001	1140.091	2751.743
	BR	BW	2113.250	448.555	0.002	848.310	3378.190

Notes. Significant *p* values are highlighted in bold.

Specifically, under the left-turn road condition, the effect of AR crash warning icon systems on the first fixation time is statistically significant ($F(2,22) = 5.793$, $p = 0.010$). The difference between the BD group (1371.167 ± 197.780 ms) and the BR group (1198.971 ± 93.630 ms) is not statistically significant ($p = 1.000$). However, the difference between the BD group and the BW group (735.667 ± 52.66 ms) is statistically significant ($p = 0.032$). The difference between the BR group and the BW group is also statistically significant ($p < 0.001$).

Under the straight road condition, the effect of AR crash warning icon systems on the first fixation time is also statistically significant ($F(2,22) = 5.208$, $p = 0.014$). However, the differences between the BD group (1136.250 ± 103.138 ms), the BR group (630.833 ± 155.850 ms), and the BW group (656.750 ± 146.791 ms) are not statistically significant ($p > 0.05$).

Under the right-turn road condition, the effect of AR crash warning icon systems on the first fixation time is also statistically significant ($F(2,22) = 18.326$, $p < 0.001$). The difference between the BD group (2562.583 ± 261.983 ms) and the BR group (2729.917 ± 387.982 ms) is not statistically significant ($p = 1.000$). However, the difference between the BD group and the BW group (616.667 ± 106.372 ms) is statistically significant ($p < 0.001$), and the difference between the BR group and the BW group is also statistically significant ($p = 0.002$).

Additionally, Figure 8b illustrates the performance of drivers' first fixation times for the three types of AR warning icon displays under different turning lane conditions.

4. Discussion

This study aimed to investigate the impact of three AR crash warning icon types (BW, BD, BR) on drivers' visual attention at different turning lanes in pedestrian-related acci-

dents. Multiple metrics, including total gaze duration, number of fixation points, and time to first fixation, indicate that the hazard-based warning icon display type (BW) significantly enhances drivers' attention to dangerous pedestrians in the driving environment compared to the other two warning icon display types (BD, BR). This optimization of drivers' visual attention allocation generally supports the H1 hypothesis, which posited significant differences in the effectiveness of various AR warning icon types for drivers' perception of potential hazards. As hypothesized, the BW-type warning icons most effectively facilitated hazard recognition. This finding aligns with the results of Calvi et al. [30], demonstrating that dynamic, location-based warnings (like our BW condition) lead to faster hazard detection times and more accurate responses compared to static or dashboard-based warnings. Similarly, the study by Lindemann et al. (2021) corroborates the conclusion that dynamic AR warnings are more effective than fixed-position warnings [31]. The results underscore the superiority of hazard-based AR warning icons in enhancing drivers' situational awareness and response to potential pedestrian-related dangers. This research contributes to the growing body of evidence supporting the implementation of dynamic, context-aware warning systems in automotive safety technology.

As hypothesized, the BW-type warning icons most effectively facilitated hazard recognition. Multiple metrics, including total gaze duration, number of fixation points, and time to first fixation, indicate that BW significantly enhances drivers' attention to dangerous pedestrians in the driving environment compared to the other two warning icon display types (BD, BR). This optimization of drivers' visual attention allocation fully aligns with hypothesis H1c, which stated that "AR warning icons based on warnings (BW) displayed at the location of dangerous pedestrians and moving with them will most effectively promote hazard recognition, being more effective than BD and BR displays". Previous research findings suggest that since the display positions of BD- and BR-type warning information do not change with the movement of dangerous pedestrians, these warning icons compete with potential hazards in the driving scene for drivers' attentional resources [20,22]. Consequently, during pedestrian-vehicle interactions, drivers' gaze duration on dangerous pedestrians is shorter, as they allocate a portion of their attention to the warning icons, resulting in decreased focus on potential risks in the scene. The superior performance of BW-type icons can be attributed to their dynamic nature, which aligns the warning with the actual location of the hazard. This alignment reduces the cognitive load on drivers, allowing them to more efficiently process and respond to potential dangers. The findings underscore the importance of context-aware, spatially-congruent warning systems in enhancing driver safety, particularly in complex urban environments with high pedestrian activity.

The research findings partially support hypothesis H2, which posited significant differences in the impact of various turning lane types on drivers' visual behavior and hazard perception. During turning maneuvers, the BW group demonstrated significantly shorter time to first fixation compared to the BD and BR groups. This indicates that the BW-type warning icon display method more effectively captures drivers' attention, enabling them to swiftly detect potential hazards and make decisions, thereby ensuring driving safety. In straight lane scenarios, no significant differences were observed in the time to first fixation across the three icon types. This suggests that in this context, drivers exhibit strong perception of dangerous pedestrians, unaffected by the warning icon type. The longer time to first fixation in the BD and BR groups implies that after the warning icons appear, drivers focus on the icons themselves, unable to promptly return their attention to the driving environment. This phenomenon, known as cognitive tunneling, aligns with previous research findings [14]. Moreover, due to the relatively fixed positions of BD and BR icons, drivers need to search for hazardous objects in the forward road environment after the icon appears. This process may lead to an extended time to first fixation on the hazardous objects for drivers in these groups.

These results underscore the context-dependent effectiveness of different AR warning icon types. The BW-type icons demonstrate superior performance in more complex driving scenarios, such as turning maneuvers, where rapid hazard detection is crucial. However, in

simpler straight-lane driving, the impact of icon type appears to be less pronounced. This nuanced understanding of icon effectiveness across various driving contexts can inform the design of more adaptive and efficient AR-HUD systems, potentially enhancing overall driving safety in diverse urban environments.

Consistent with the initial hypotheses, significant differences were observed in the eye movement data of the BD and BR groups across various driving lane conditions. This indicates that under BD and BR conditions, the visual field differences caused by turning lanes affect drivers' perception of dangerous pedestrians. In other words, under these two types of warning icon conditions, drivers are influenced by the vehicle's driving lane. Moreover, executing turning maneuvers increases the complexity of the driving task. Drivers must respond to the current traffic situation of the turning road, rather than simply maintaining longitudinal and lateral vehicle control [32]. Our research findings partially support Wang et al.'s observations, particularly regarding the potential for BD- and BR-type warnings to cause distraction [33]. However, our study further demonstrates that BW-type warnings maintain consistent effectiveness across different driving scenarios, forming an interesting contrast with Wang et al.'s research. This discrepancy may stem from our implementation of dynamic, context-aware AR warning designs, which appear to better balance information presentation and attention management. The BW-type icons, by adapting to the driving environment and potential hazards, seem to mitigate the lane-dependent effects observed with the BD and BR types. This adaptive approach potentially reduces cognitive load on drivers, allowing for more efficient processing of critical safety information regardless of the specific driving lane or maneuver being executed. These findings underscore the importance of considering the interaction between warning icon design and driving context in the development of AR-HUD systems. The superior performance of the BW-type icons across various driving scenarios suggests that future AR warning systems should prioritize dynamic, hazard-based designs to enhance driver safety and situational awareness in complex urban environments.

In this study, drivers demonstrated a higher ability to detect hazardous objects and maintain increased attention on straight roads without turning tasks compared to left-turn and right-turn lanes. For the based on warning (BW) warning icon, no significant differences were observed in drivers' eye movement data across various driving lanes. This suggests that under BW conditions, driver performance remains consistent regardless of the lane, with relatively stable behavior across different driving lanes and minimal variations due to differences in visual fields during turning maneuvers. Although no significant differences were found in subjective satisfaction ratings among the three icon types, the BW icon received higher satisfaction scores compared to the other two icon types.

While the results have addressed the research objectives, this research still has limitations. The participant pool in this study was relatively homogeneous, primarily consisting of young adults with limited driving experience. Consequently, the findings may only represent characteristics specific to this demographic group. Research indicates that as drivers age, their abilities in attention allocation, environmental awareness, information processing, and decision-making may gradually decline [27,34,35]. Conversely, extensive driving experience can enhance drivers' ability to perceive hazardous objects [28,36–38], potentially reducing collision rates. Future research should explore the applicability of augmented reality (AR) icons across drivers of varying ages, driving experiences, and cognitive levels. This expanded scope would provide a more comprehensive understanding of the effectiveness of AR icons in diverse driver populations.

This study focused solely on emergency situations involving vehicle-pedestrian accidents. However, real-world driving environments are considerably more complex, with various hazardous objects exhibiting different velocities, motion states, spatial positions, and other characteristics. To optimize the application of driver assistance systems in promoting driving safety more effectively, future research should consider a more nuanced categorization of accident types. Subsequent studies could explore the applicability of augmented reality (AR) warning information displays across diverse accident scenarios.

This approach would provide a more comprehensive understanding of how AR warning systems perform in various hazardous situations, potentially leading to more tailored and effective safety interventions.

Despite the use of virtual reality technology for the driving simulation, particularly the implementation of 360° video to enhance ecological validity [39], the ecological validity of simulated experiments remains limited. The study's conclusions should be interpreted cautiously, as the driving experience in virtual environments may differ significantly from real-world driving conditions. These differences could potentially influence drivers' responses and behaviors, thereby affecting the external validity of the research findings. The disparity between simulated and real-world driving environments may impact various aspects of driver performance, including perception, decision-making, and physical responses. Consequently, while the virtual reality simulation provides valuable insights, it is crucial to acknowledge its limitations in fully replicating the complexities and nuances of actual driving scenarios.

An additional limitation of this study lies in its experimental design, which only considered scenarios with a single warning sign. In real-world driving environments, drivers frequently encounter multiple potential hazards simultaneously, necessitating the ability to process and respond to concurrent warning signals. The simultaneous presentation of multiple warnings may compete for drivers' attentional resources, potentially affecting their capacity to effectively identify, prioritize, and react to the most urgent dangers. This simplification in the experimental design may limit the generalizability of the findings to more complex driving scenarios. Real-world driving often requires drivers to manage a multitude of information inputs, including various warning signals, traffic conditions, and environmental factors. The cognitive load associated with processing multiple warnings could significantly influence drivers' performance and decision-making processes in ways not captured by this study's single-warning paradigm.

Future research should further explore the impact on drivers of simultaneously presenting multiple warning signals. This exploration should include, but not be limited to, the following aspects: (a) Expanding the range of study participants: Investigate the responses of drivers with varying ages, driving experiences, and cognitive levels to augmented reality (AR) icons. This approach would enhance the generalizability of research findings. (b) Examining the applicability of AR warning information displays across diverse accident scenarios, such as vehicle collisions and sudden obstacles. (c) Designing experiments that more closely replicate real-world driving conditions to improve ecological validity. This could involve using advanced driving simulators or conducting real-vehicle tests. (d) Investigating the effects of concurrent multiple warning signals on drivers' performance and cognitive load. (e) Conducting similar studies in different countries and cultural contexts to explore the influence of cultural factors on the effectiveness of AR warning systems.

These research directions would address the current study's limitations and provide a more comprehensive understanding of AR warning systems in diverse driving scenarios. By encompassing a wider range of variables and conditions, future studies can contribute to the development of more effective and universally applicable driver assistance technologies.

5. Conclusions

Efficient design and development of HUDs is essential for improving drivers' detection and perception abilities on the road, ultimately enhancing safety and driving performance. While the integration of AR with HUDs has addressed some limitations of traditional systems, the lack of unified design standards in the commercial HUD industry remains a challenge. The proliferation of complex and diverse HUD icon design, coupled with significant variations among HUD systems, perpetuates the issue of driver distraction associated with traditional in-vehicle displays. Building upon previous research on HUD information presentation, this study incorporates AR icon characteristics to classify AR crash warning icon systems. Findings from this investigation aim to provide design guidelines for AR

crash warning icon systems, with the goal of minimizing driver distraction and enhancing the detection of potential hazards in pedestrian-vehicle scenarios.

The findings reveal that AR crash warning icon systems based on hazardous objects enable drivers to identify dangerous pedestrians in the environment more rapidly. Moreover, this type of AR icon demonstrates consistent visual behavior across various turning lanes, unaffected by differences in field of view, indicating strong applicability. In contrast, screen-fixed AR crash warning icon systems may divert drivers' attention, competing with potential risks in the environment for limited visual resources.

These findings specifically indicate that when designing AR crash warning icon systems, priority should be given to the dynamic tracking capability of the icons. This ensures that icons can move and update with hazardous objects, helping drivers identify and respond to potential threats in the driving environment accurately and promptly. Furthermore, the study emphasizes avoiding the use of screen-fixed icon types to reduce competition for visual resources and minimize distraction, thereby enhancing overall safety during driving.

Future research should explore the impact of multiple concurrent warning signals on drivers, including the following: (a) Investigating responses of diverse driver populations (age, experience, cognitive levels) to AR icons, enhancing generalizability. (b) Examining AR warning displays across various accident scenarios. (c) Designing experiments that closely replicate real-world driving for improved ecological validity. (d) Studying the effects of multiple warning signals on driver performance and cognitive load. These directions would address limitations and provide a comprehensive understanding of AR warning systems in diverse driving scenarios, contributing to the development of effective, universally applicable driver assistance technologies.

Author Contributions: Conceptualization, methodology, validation, review and editing, Z.W.; investigation, data curation, formal analysis, writing—original draft preparation, Y.L.; visualization and editing, G.L.; supervision, funding acquisition, X.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by Social Science Foundation of Fujian province, China [Grant number FJ2021B178], Humanities and Social Science Foundation, the Ministry of Education of China [Grant number 22YJA760001], National Natural Science Foundation of China [Grant number 51808232], Provincial Training Program of Innovation and Entrepreneurship for Undergraduates [Grant number S202310385048].

Informed Consent Statement: Written informed consent has been obtained from the participants to publish this paper.

Data Availability Statement: Data is contained within the article.

Conflicts of Interest: The authors declare no conflict of interest.

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